إعداد الطالبة :- بتول هاتف جادر

مصفوفة ثنائية البعد

حل تمارين بدون استخدام القارئ الScanner

JavaApplication4 - NetBeans IDE 8.2

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Start Page | JavaApplication4.java

Source | History

```java
package javaapplication4;
public class JavaApplication4 {
    public static void main(String[] args) {
        int x[][]={
            {1,2,3},
            {2,3,4},
            {4,5,6}
        };
      System.out.println("output 2D Arrayes : ");
      for(int i=0;i<3;i++){
      for(int j=0;j<3;j++)
      System.out.print(x[i][j]+" ");
      System.out.println();
      }
    }}
```

Output - JavaApplication4 (run)

```
run:
output 2D Arrayes :
1 2 3
2 3 4
4 5 6
BUILD SUCCESSFUL (total time: 0 seconds)
```

طباعة عناصر مصفوفة 2D Array

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Source | History

```java
package javaapplication4;
public class JavaApplication4 {
    public static void main(String[] args) {
        int x[][]={
            {1,2,3},
            {2,3,4},
            {4,5,6}
        };
      int sum=0;
      System.out.println("output 2D Arrayes : ");
      for(int i=0;i<3;i++){
      for(int j=0;j<3;j++)
      System.out.print(x[i][j]+" ");
      System.out.println();
      }
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
      sum=sum+x[i][j];
    }
      System.out.println("sum is :"+sum);
    }}
```

Output - JavaApplication4 (run)

```
run:
output 2D Arrayes :
1 2 3
2 3 4
4 5 6
sum is :30
BUILD SUCCESSFUL (total time: 0 seconds)
```

✓ طباعة عناصر مصفوفة

2D Array

✓ وايجاد جمع المصفوفة

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JavaApplication4 - NetBeans IDE 8.2

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Start Page | JavaApplication4.java

```java
package javaapplication4;
public class JavaApplication4 {
    public static void main(String[] args) {
        int x[][]={
            {1,2,3},
            {2,3,4},
            {4,5,6}
        };
      int sum=0;
      System.out.println("output 2D Arrayes : ");
      for(int i=0;i<3;i++){
      for(int j=0;j<3;j++)
      System.out.print(x[i][j]+" ");
      System.out.println();
      }
      for(int i=0;i<3;i++){
      for(int j=0;j<3;j++)
        sum=sum+x[i][j];
      }
      System.out.println("sum is :"+sum);
       System.out.println("sqrt is :"+Math.sqrt(sum));
    }}
```

Output - JavaApplication4 (run)

```
run:
output 2D Arrayes :
1 2 3
2 3 4
4 5 6
sum is :30
sqrt is :5.477225575051661
BUILD SUCCESSFUL (total time: 0 seconds)
```

- ✓ طباعة عناصر مصفوفة 2D Array
- ✓ وايجاد جمع المصفوفة
- ✓ وايجاد الجذر التربيعي للمجموع

JavaApplication4 - NetBeans IDE 8.2

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Source | History

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public class JavaApplication4 {
    public static void main(String[] args) {
        int x[][]={
            {1,2,3},
            {2,3,4},
            {4,5,6}
        };
      int sum=0;
      System.out.println("output 2D Arrayes : ");
      for(int i=0;i<3;i++){
      for(int j=0;j<3;j++)
      System.out.print(x[i][j]+" ");
      System.out.println();
      }
      for(int i=0;i<3;i++){
      for(int j=0;j<3;j++)
        sum=sum+x[i][j];
      }
      System.out.println("sum is :"+sum);
       System.out.println("sqrt is :"+Math.sqrt(sum));
         System.out.println("avarage is :"+sum/9);
    }}
```

Output - JavaApplication4 (run)

```
run:
output 2D Arrayes :
1 2 3
2 3 4
4 5 6
sum is :30
sqrt is :5.477225575051661
avarage is :3
BUILD SUCCESSFUL (total time: 0 seconds)
```

- ✓ طباعة عناصر مصفوفة 2D Array
- ✓ وايجاد جمع المصفوفة
- ✓ وايجاد الجذر التربيعي للمجموع
- ✓ وايجاد المعدل لعناصر المصفوفة

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Source | History

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        int x[][]={
            {1,2,3},
            {2,3,4},
            {4,5,6}
        };
      System.out.println("output 2D Arrayes : ");
      for(int i=0;i<3;i++){
      for(int j=0;j<3;j++)
      System.out.print(x[i][j]+" ");
      System.out.println();
      }
    int max=x[0][0];
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++){
        if(max<x[i][j])
           max=x[i][j];
    }}
    System.out.println("max number is: "+ max);
    }}
```

Output - JavaApplication4 (run)

```
run:
output 2D Arrayes :
1 2 3
2 3 4
4 5 6
max number is: 6
BUILD SUCCESSFUL (total time: 0 seconds)
```

✓ طباعة عناصر مصفوفة 2D Array

✓ وايجاد أكبر عنصر للمصفوفة

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Source | History

```java
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public class JavaApplication4 {
    public static void main(String[] args) {
        int x[][]={
            {1,2,3},
            {2,3,4},
            {4,5,6}
        };
     System.out.println("output 2D Arrayes : ");
     for(int i=0;i<3;i++){
     for(int j=0;j<3;j++)
     System.out.print(x[i][j]+" ");
     System.out.println();
     }
    int min=x[0][0];
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++){
        if(min>x[i][j])
           min=x[i][j];
    }}
    System.out.println("min number is: "+ min);
    }}
```

Output - JavaApplication4 (run)

```
run:
output 2D Arrayes :
1 2 3
2 3 4
4 5 6
min number is: 1
BUILD SUCCESSFUL (total time: 0 seconds)
```

✓ طباعة عناصر مصفوفة 2D Array

✓ وايجاد أصغر عنصر للمصفوفة

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package javaapplication4;
public class JavaApplication4 {
    public static void main(String[] args) {
        int x[][]={
            {1,2,3},
            {2,3,4},
            {4,5,6}
        };
     System.out.println("output 2D Arrayes : ");
     for(int i=0;i<3;i++){
     for(int j=0;j<3;j++)
     System.out.print(x[i][j]+" ");
     System.out.println();
     }
    int min=x[0][0],max=x[0][0];
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++){
        if(min>x[i][j])
           min=x[i][j];
          if(max<x[i][j])
             max=x[i][j];
    }}
    System.out.println("min number is: "+ min);
    System.out.println("max number is: "+ max);
    }}
```

Output - JavaApplication4 (run)

```
run:
output 2D Arrayes :
1 2 3
2 3 4
4 5 6
min number is: 1
max number is: 6
BUILD SUCCESSFUL (total time: 0 seconds)
```

- ✓ طباعة عناصر مصفوفة 2D Array
- ✓ وايجاد أصغر عنصر للمصفوفة
- ✓ وايجاد أكبر عنصر للمصفوفة

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JavaApplication4 - NetBeans IDE 8.2

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Source | History

```java
package javaapplication4;
public class JavaApplication4 {
    public static void main(String[] args) {
        int x[][]={
            {1,2,3},
            {2,3,4},
            {4,5,6}
        };
        System.out.println("output 2D Arrayes : ");
        for(int i=0;i<3;i++){
        for(int j=0;j<3;j++)
        System.out.print(x[i][j]+" ");
        System.out.println();
        }
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
        if(i==j)
        System.out.println(x[i][j]);
    else
        System.out.print(" ");
        System.out.println("");
    }
    }}
```

Output - JavaApplication4 (run)

```
run:
output 2D Arrayes :
1 2 3
2 3 4
4 5 6
1

 3

  6

BUILD SUCCESSFUL (total time: 0 seconds)
```

✓ طباعة عناصر مصفوفة 2D Array

✓ طباعة القطر الرئيسي فقط

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JavaApplication4 - NetBeans IDE 8.2

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  - Libraries

Start Page | JavaApplication4.java

Source | History

```java
package javaapplication4;
public class JavaApplication4 {
    public static void main(String[] args) {
        int x[][]={
            {1,2,3},
            {2,3,4},
            {4,5,6}
        };
     System.out.println("output 2D Arrayes : ");
     for(int i=0;i<3;i++){
     for(int j=0;j<3;j++)
     System.out.print(x[i][j]+" ");
     System.out.println();
     }
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
        if(i+j==2)
        System.out.println(x[i][j]);
    else
        System.out.print(" ");
        System.out.println("");
    }
    }}
```

Output - JavaApplication4 (run)

```
run:
output 2D Arrayes :
1 2 3
2 3 4
4 5 6
  3

 3

4

BUILD SUCCESSFUL (total time: 0 seconds)
```

✓ طباعة عناصر مصفوفة

2D Array

✓ طباعة القطر الثانوي فقط

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JavaApplication6 - NetBeans IDE 8.2

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Start Page | JavaApplication4.java | JavaApplication6.java

```java
package javaapplication6;
public class JavaApplication6 {
    public static void main(String[] args) {
            int x[][]={
            {1,2,3,2},
            {2,3,4,7},
            {4,5,6,8} ,
            {4,5,6,3}
        };
     System.out.println("output 2D Arrayes : ");
     for(int i=0;i<4;i++){
     for(int j=0;j<4;j++)
     System.out.print(x[i][j]+" ");
     System.out.println();
     }
    for(int i=0;i<4;i++){
    for(int j=0;j<4;j++)
        if(i==j)
        System.out.print(x[i][j]+" ");
       else if(i+j==3)
      System.out.print(x[i][j]);
       else
        System.out.print(" ");
        System.out.println(" ");
    }
    }}
```

Output - JavaApplication6 (run)

```
run:
output 2D Arrayes :
1 2 3 2
2 3 4 7
4 5 6 8
4 5 6 3
1   2
 3 4
 56
4  3
BUILD SUCCESSFUL (total time: 0 seconds)
```

✓ طباعة عناصر مصفوفة

2D Array

✓ طباعة القطر الرئيسي والثانوي

JavaApplication6 - NetBeans IDE 8.2

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Start Page | JavaApplication4.java | JavaApplication6.java

Source | History

```java
package javaapplication6;
public class JavaApplication6 {
    public static void main(String[] args) {
            int x[][]={
            {1,2,3,2},
            {2,3,4,7},
            {4,5,6,8} ,
            {4,5,6,3}
        };
      System.out.println("output 2D Arrayes : ");
          for(int i=0;i<4;i++){
        for(int j=0;j<4;j++)
      System.out.print(x[i][j]+" ");
      System.out.println();}
      System.out.println("----------------");
      for(int i=0;i<4;i++){
      for(int j=0;j<4;j++)
          if(i==j){
          x[i][j]=1;
          }}
       for(int i=0;i<4;i++){
      for(int j=0;j<4;j++)
      System.out.print(x[i][j]+" ");
      System.out.println();
      }}
    }
```

Output - JavaApplication6 (run)

```
run:
output 2D Arrayes :
1 2 3 2
2 3 4 7
4 5 6 8
4 5 6 3
----------------
1 2 3 2
2 1 4 7
4 5 1 8
4 5 6 1
BUILD SUCCESSFUL (total time: 0 seconds)
```

✓ طباعة عناصر مصفوفة 2D Array

✓ استبدال القطر الرئيسي بال 1

JavaApplication6 - NetBeans IDE 8.2

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Start Page | JavaApplication4.java | JavaApplication6.java

Source | History

```java
package javaapplication6;
public class JavaApplication6 {
    public static void main(String[] args) {
            int x[][]={
            {1,2,3,2},
            {2,3,4,7},
            {4,5,6,8} ,
            {4,5,6,3}
        };
      System.out.println("output 2D Arrayes : ");
          for(int i=0;i<4;i++){
        for(int j=0;j<4;j++)
      System.out.print(x[i][j]+" ");
      System.out.println();}
      System.out.println("-----------------");
      for(int i=0;i<4;i++){
      for(int j=0;j<4;j++)
          if(i+j==3){
          x[i][j]=1;
          }}
       for(int i=0;i<4;i++){
      for(int j=0;j<4;j++)
      System.out.print(x[i][j]+" ");
      System.out.println();
      }}
    }
```

Output - JavaApplication6 (run)

```
run:
output 2D Arrayes :
1 2 3 2
2 3 4 7
4 5 6 8
4 5 6 3
-----------------
1 2 3 1
2 3 1 7
4 1 6 8
1 5 6 3
BUILD SUCCESSFUL (total time: 0 seconds)
```

✓ طباعة عناصر مصفوفة 2D Array

✓ استبدال القطر الثانوي بال 1

```java
package javaapplication34;
public class JavaApplication34 {
    public static void main(String[] args) {
int[][] x={{1,-2,2},{-1,-3,2}};
System.out.println("nagitive");
for(int i=0;i<x.length;i++){
    for(int j=0;j<x[i].length;j++){
    if(x[i][j]<0){
   System.out.println(x[i][j]);
    }}}
    }}
```

```
run:
nagitive
-2
-1
-3
BUILD SUCCESSFUL (total time: 0 seconds)
```

طباعة اعداد سالبة من مصفوفة ثنائية البعد